

American Powerhouses: Economic Dominance within State Lines

GROUP 25

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Our project revolves around the individual states within our nation and their economic performance over time. It's very interesting to us how certain states have really stood above the rest over the past few decades and we want to know why that is. Two of us are economics majors, and the other a government major so we are all naturally concerned about the overarching mechanisms that govern our daily life. The world is composed of vast systems that dictate how we live and breathe, and the individual states within the U.S are a big part of that in relation to our lives. That's why our research will focus on the idea that the economic growth of U.S states are tied to the largest companies, population growth, and R&D spending within their borders. Our hypothesis is that the frequency of industry leaders, specifically Fortune 500 companies in addition to population growth and increases in R&D spending have largely contributed to why certain states top the economic leaderboard in terms of GDP.

In the creation of this proposal we looked for various academic papers that could give us a deeper understanding of what we want to learn about. Through Google Scholar, we found several that provided some useful insights. One such example is Lee et al. (2013) who found that "big businesses," or those in various databases like the Fortune 500 drove significant and positive economic growth and were associated with stability of such growth in countries across the world. Jorgenson & Stiroh (2000) found that high tech industries were the biggest drivers of regional growth in the second half of the 20th century, while others industries lagged behind. Both of these papers corroborate our hypothesis because they each found a positive correlation between industry leaders and the economic performance of where they were located.

Although it wasn't something we had considered initially, when looking for articles about U.S state growth several popped up that discussed the importance of research and development. Blanco & Prieger (2015) found that R&D has a massive effect on state economic performance in the long run. This spreads to other states as well, with them finding 70-80% of returns spilling over. Similarly, Glaeser et al. (1992) found that local competition and a variety of industries had a significant effect on regional development when they examined why some cities grew faster than others. They found that economic growth was closely related to knowledge spillover which is when innovations and ideas happen across different industries, benefiting from increased productivity and long term economic growth. These papers give us insight about how R&D creates positive economic returns for individual states which create a domino effect on its

neighbors as well. This gives us more support for our hypothesis because we can infer from our findings that R&D is a big factor in why certain states have performed better than others.

How we want to test our hypothesis is by finding a correlation between R&D spending, population growth, and quantity of industry leaders with annual state economic growth. We plan on utilizing the databases we've acquired from Gigasheet, The Fed Bank of St. Louis, and National Center for Science and Engineering Statistics in order to find out for ourselves if the data backs it up. We already have the research background to give us an academic basis for our question, but now our findings will test the validity of our claim.

References

Blanco, L.R., Gu, J. and Prieger, J.E. (2016), The Impact of Research and Development on Economic Growth and Productivity in the U.S. States. *Southern Economic Journal*, 82: 914-934. <https://doi.org/10.1002/soej.12107>

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Jorgenson, D. W., & Stiroh, K. J. (2000). U.S. Economic Growth at the Industry Level. *American Economic Review*, 90(2), 161–167. American Economic Association
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Lee, K., Kim, B.-Y., Park, Y.-Y., & Sanidas, E. (2013). Big businesses and economic growth: Identifying a binding constraint for growth with country panel analysis. *Journal of Comparative Economics*, 41(2), 561–582. REPEC <https://doi.org/10.1016/j.jce.2012.07.006>

Databases

Annual State GDP from the Federal Reserve of St. Louis Bank (links to Texas specifically, but you can find each state through the search bar)
<https://fred.stlouisfed.org/series/TXNGSP>

Annual State Population from the Federal Reserve of St. Louis Bank (same as above)
<https://fred.stlouisfed.org/series/TXPOP>

Fortune 500: List of Fortune 500 companies and their attributes from Fortune magazine.
<https://www.gigasheet.com/sample-data/fortune-500-companies>

Annual State R&D Spending from National Center for Science and Engineering Statistics
<https://nces.nsf.gov/surveys/state-government-research-development/2024>